

RAJALAKSHMI ENGINEERING COLLEGE

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
CS23334 – FUNDAMENTALS OF DATA SCIENCE LAB (REGULATION 2023)

Experiment No.	20
Experiment Name	Understand Data Preprocessing in Data Science

DESCRIPTION

To perform the experiment: Understand Data Preprocessing in Data Science.

CODE

```
import pandas as pd
from sklearn.preprocessing import StandardScaler, LabelEncoder

df = pd.DataFrame({
    "Country": ["France", "Spain", "Germany", "Spain", None],
    "Age": [44, 27, 30, None, 40],
    "Salary": [72000, 48000, 54000, 61000, None],
    "Purchased": ["No", "Yes", "No", "No", "Yes"]
})
print("Original data:")
print(df)
df["Country"] = df["Country"].fillna(df["Country"].mode()[0])
df["Age"] = df["Age"].fillna(df["Age"].median())
df["Salary"] = df["Salary"].fillna(df["Salary"].mean())
encoder = LabelEncoder()
df["Country_Code"] = encoder.fit_transform(df["Country"])
df["Purchased_Code"] = encoder.fit_transform(df["Purchased"])
df[["Age_Scaled", "Salary_Scaled"]] = StandardScaler().fit_transform(
    df[["Age", "Salary"]])
print("\nPreprocessed data:")
print(df)
```

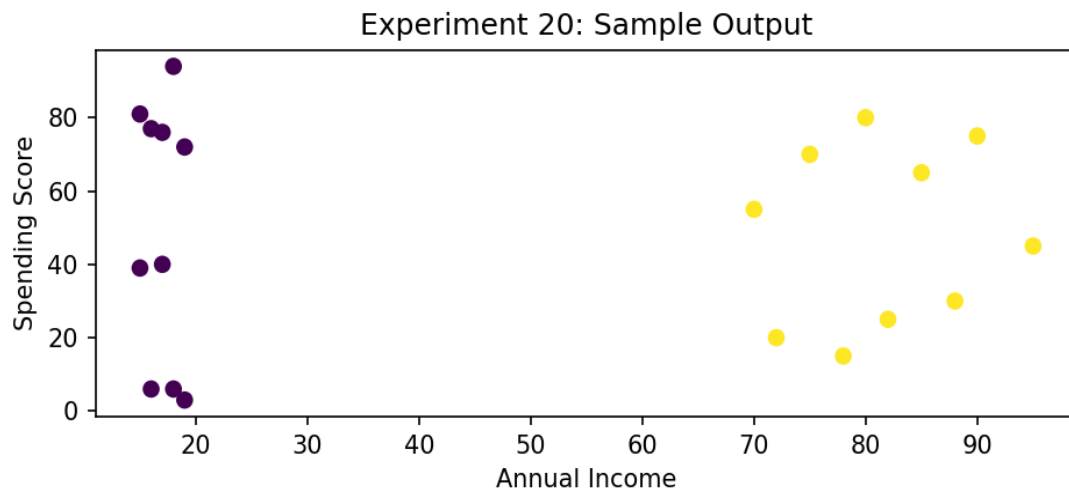
SAMPLE INPUT

Sample input: country, age, salary and purchased fields, with some missing values.

OUTPUT — SAMPLE CONSOLE OUTPUT

Preprocessed table includes imputed values, category codes and scaled numerical features.

OUTPUT DIAGRAM / GRAPH



RESULT

The program demonstrates the stated experiment using the sample data. Verify the output by running the code in your Python environment before academic submission.

Student Signature

Faculty Signature